TRANSFORMER SUMMARIZER — COMPLETE EVALUATION PIPELINE

One article | 3 models | step-by-step transformations | ROUGE-1/2/L | NER P/R/F1 | Sentiment LR | Topic RF

STAGE 1

STAGE 1 — INPUT ARTICLE (CNN/DailyMail dataset)

ARTICLE (same example used in ALL stages):

"Scientists discovered a massive black hole located near Earth,
raising new questions about galactic formation. The team, led by Dr. Sarah Chen at MIT,
used the James Webb Space Telescope. Mass: 10 billion x Sun. Formed in early universe."

CORPUS	SIZE	AVG LEN	VOCAB	SPLIT
CNN/DailyMail	280,327 arts	~780 tokens	~50k BPE	train/val/test

REFERENCE SUMMARY

Gold standard (human-written):

"Scientists found a huge black
hole close to Earth. MIT team
led by Dr Chen used Webb."

ROUGE + NER scoring only

(no ref for Sentiment/Topic)

extractive

abstractive (sequential)

abstractive (parallel)

STAGE 2

(1) TF-IDF EXTRACTIVE

APPROACH: Extractive — copies full sentences verbatim

No neural network. Scores each sentence by keyword
importance (TF-IDF) then copies top-K as the summary.

Key formulas:

TF(w,s) = count(w,s) / |s| IDF(w) = log(N/df(w))

TF-IDF(w,s) = TF x IDF   score(s) = SUM TF-IDF

TRANSFORMATION TRACE

INPUT: Article Sentence S1

"Scientists discovered a massive black hole near Earth."

STEP 1 — TOKENIZE + REMOVE STOPS |S1| = 8 content words

[scientists, discovered, massive, black, hole, near, earth] +1 extra

STEP 2 — TERM FREQ TF = 1/8 = 0.125 (each word appears once)

STEP 3 — IDF = log(N/df) N = 280,327

Word	df	IDF	
scientists	1,200	5.45	<- rare, informative
discovered	8,500	3.50	
massive	4,200	4.20	
black	12,000	3.15	<- common, low IDF
hole	800	5.86	<- very rare, HIGH
near	15,000	2.93	
earth	3,000	4.54	

STEP 4 — TF-IDF = 0.125 x IDF

scientists=0.681 discovered=0.438 massive=0.525

hole=0.733 (highest) near=0.366 earth=0.568

STEP 5 — SENTENCE SCORE = SUM TF-IDF

score(S1) = 4.175 -> RANK 1 SELECTED

score(S2) = 3.680 -> RANK 2 SELECTED [Dr Chen, MIT, Webb]

score(S3) = 2.140 -> RANK 3 not selected

STEP 6 — OUTPUT: copy S1, S2 verbatim

"Scientists discovered a massive black hole..."

"discovered" kept; ref uses "found" -> ROUGE miss

ROUGE-1: 17.0% ROUGE-2: 6.1% ROUGE-L: 9.2%

(2) BiLSTM + BAHDANAU

APPROACH: Abstractive — BiLSTM encoder-decoder

Bidirectional LSTM + Bahdanau single-head attention.

Key:

h_i=[LSTM_f(x_i,h_{i-1})|LSTM_b(x_i,h_{i+1})] 512-dim

e(t,i)=v*tanh(W_s*s_t+W_h*h_i) a=softmax(e) c=sum(a*h)

TRANSFORMATION TRACE

ENCODER — BiLSTM processes 8 tokens

h1=[LSTM_f(x1,h0)|LSTM_b(x1,h8)]=[0.82,-0.21,...,-0.31,0.73,...] 512d

h2=LSTM_f(x2,h1) SEQUENTIAL (needs h1) h2=[0.71,-0.34,...] 512d

DECODER STEP 1 — Bahdanau attention

e(1,1)=0.72 "scientists" e(1,2)=0.68 "discovered"

e(1,3)=0.31 "massive" e(1,4)=0.28 "black"

a=[0.38, 0.35, 0.12, 0.08,...] scientists+discovered dominate

c1=0.38*h1+0.35*h2+0.12*h3+... (512-dim context)

P(researchers|s1,c1)=0.34 P(scientists)=0.21 P(team)=0.18

-> Emit: "researchers" (paraphrase of "scientists"!)

DECODER STEP 2 — recompute attention

e(2,2)=0.81 "discovered" focus! a2=[0.15,0.43,0.18,...]

P(found|s2,c2)=0.41 -> "found" (paraphrase of "discovered"!)

Sequential: step t needs s_{t-1} -> no parallelism

Long-range: context decays over many steps

OUTPUT: "researchers found large black hole near earth..."

ROUGE-1: ~20% ROUGE-2: ~8% ROUGE-L: ~14% (approx)

(3) TRANSFORMER Seq2Seq

APPROACH: Abstractive — Transformer Seq2Seq

Pure attention, no recurrence. All positions in parallel.

Key:

Q=X*W_Q K=X*W_K V=X*W_V d_model=512, 4 heads, d_k=128

Attn=softmax(Q*Kt/sqrt(128))*V MultiHead=Concat(h1..h4)*W_O

TRANSFORMATION TRACE

INPUT: BPE + positional encoding

[scientists,discov-ered,a,massive,black,hole,near,earth] pos 1..8

X[i]=embed(token_i)+PE(i) PE(1,0)=sin(1)=0.841 512-dim

MULTI-HEAD ATTENTION — ALL 8 POSITIONS PARALLEL

HEAD 1 SYNTACTIC: score[scientists,discov-ered]=4.2 [black,hole]=4.5

HEAD 2 SEMANTIC : score[discov-ered,found]=3.8 <- PARAPHRASE!

score[massive,huge]=3.6 score[near,close]=3.4

HEAD 3 CO-REF : score[MIT,team]=3.9 score[Chen,she]=3.7

HEAD 4 N-GRAM : score[James,Webb]=4.3 score[black,hole]=4.5

MultiHead=Concat(h1,h2,h3,h4)*W_O [8x512] ALL PARALLEL!

DECODER — cross-attention to encoder outputs

Pos	Attends to	Generates	
1:	scientists(0.82)	-> "Scientists"	
2:	discov-ered(0.79)	-> "found"	<- HEAD 2 synonym!
3:	a(0.91)	-> "a"	
4:	massive(0.77)	-> "huge"	<- HEAD 2 synonym!
5:	black(0.88)	-> "black"	
6:	hole(0.92)	-> "hole"	
7:	near(0.81)	-> "close"	<- HEAD 2 synonym!

OUTPUT: "Scientists found a huge black hole close to Earth."

ref -> ROUGE,NER

ROUGE-1: 24.5% ROUGE-2: 9.8% ROUGE-L: 19.9% BEST

STAGE 3

TF-IDF EXTRACTIVE SUMMARY

"Scientists discovered a massive black hole located near Earth."

"The team, led by Dr. Sarah Chen at MIT, used James Webb..."

-> 2 top-scoring sentences copied VERBATIM

-> "discovered" kept; ref uses "found" -> ROUGE miss

-> All entities preserved: Sarah Chen, MIT, Earth, Webb

BiLSTM SUMMARY

"researchers found large black hole near earth."

"dr chen at mit led study using webb telescope."

-> "found" paraphrase -> ROUGE match with reference!

-> Full lowercase -> NER misses most entities

-> "large" vs "huge" -> partial ROUGE miss

TRANSFORMER SUMMARY

"Scientists found a huge black hole close to Earth."

"MIT team led by Dr Chen used James Webb telescope."

-> discovered->found, massive->huge, near->close (HEAD 2)

-> Near-perfect match to reference -> high ROUGE

-> BPE lowercase: "earth" not "Earth" -> NER casing issue

Each summary enters all 4 evaluation modules independently

STAGE 4

(1) N-GRAM OVERLAP [INTRINSIC — ROUGE]

NEEDS REFERENCE

REFERENCE R (gold standard, lowercase tokens)

R = [scientists, found, a, huge, black, hole, close, to, earth, mit,
team, led, by, dr, chen, used, james, webb, telescope] |R| = 19

THREE SUMMARIES (all lowercase for comparison)

S1(TF-IDF): [scientists,discovered,a,massive,black,hole,located,near,
earth,team,led,by,dr,sarah,chen,at,mit,used,james,webb,space,telescope] |S1|=23

S2(BiLSTM): [researchers,found,large,black,hole,near,earth,dr,chen,
mit,led,study,using,webb,telescope] |S2| = 15

S3(Trans) : [scientists,found,a,huge,black,hole,close,to,earth,mit,
team,led,by,dr,chen,used,james,webb,telescope] |S3| = 19

ROUGE-1 — UNIGRAM OVERLAP

clip(w) = min(count in S, count in R) P = sum_clip / |S| R = sum_clip / |R|

TF-IDF (S1) vs R:

Matched: scientists,a,black,hole,earth,team,led,by,dr,chen,mit,used,james,webb,telescope = 15

Missed : discovered(R=found) massive(R=huge) located(R=close) sarah space = 5

clip=15 P=15/23=0.652 R=15/19=0.789 F1=0.714

BiLSTM (S2) vs R:

Matched: found,black,hole,earth,dr,chen,mit,led,webb,telescope = 10

clip=10 P=10/15=0.667 R=10/19=0.526 F1=0.588

Transformer (S3) vs R:

All 19 tokens match R exactly clip=19 P=1.000 R=1.000 F1=1.000

ROUGE-2 — BIGRAM OVERLAP

Bigram = consecutive word pair. |R_bigrams| = 18 pairs

TF-IDF bigrams: R has (scientists,found)(found,a)(a,huge)...

S1 has (scientists,discovered)(discovered,a)(a,massive)...

Match: (black,hole)(team,led)(led,by)(by,dr)(used,james)(james,webb) = 6

P=6/22=0.273 R=6/18=0.333 F1=0.300

"discovered" breaks all bigrams around it vs reference "found"

Transformer: 18/18 bigrams match -> F1=1.000 (Real corpus: 9.8%)

ROUGE-L — LONGEST COMMON SUBSEQUENCE (LCS)

LCS = longest matching subsequence IN ORDER (gaps allowed, not contiguous)

TF-IDF LCS alignment:

R : [sc, fo, a, hu, bl, ho, cl, to, ea, mi, te, le,by,dr,ch,us,ja,we,tel]

S1: [sc, -, a, -, bl, ho, -, -, ea, -, te, le,by,dr,ch, -, us,ja,we, -,tel]

LCS=[scientists,a,black,hole,earth,team,led,by,dr,chen,used,james,webb,telescope] len=14

P_L=14/23=0.609 R_L=14/19=0.737 F_L=0.667

TF-IDF copies ARTICLE word order != REFERENCE order -> short LCS

Transformer: 19-token match -> F_L=1.000 (Real corpus: 19.9%, +117% vs TF-IDF)

FINAL COMPARISON (this example / real corpus avg in brackets)

Metric	TF-IDF	BiLSTM	Transformer
ROUGE-1 F1	0.714 [17.0%]	0.588 [~20%]	1.000 [24.5%]
ROUGE-2 F1	0.300 [6.1%]	--- [~8%]	1.000 [9.8%]
ROUGE-L F1	0.667 [9.2%]	--- [~14%]	1.000 [19.9%]

Mini-example scores are high by design; real corpus has more variation.

(2) NER ENTITY PRESERVATION [INTRINSIC]

NEEDS ARTICLE

TOOLS: spaCy en_core_web_sm (rule-based) + fallback regex

Entity types tracked: PERSON ORG GPE FACILITY DATE NORP

STEP 1 — EXTRACT ARTICLE ENTITY SET A

Run spaCy on article: "...Dr. Sarah Chen at MIT...James Webb...Earth..."

Sarah Chen	[PERSON]	<- researcher name
MIT	[ORG]	<- institution
Earth	[GPE]	<- geo-political entity
James Webb Space Tel	[FAC]	<- telescope name

A = {Sarah Chen, MIT, Earth, James Webb Space Tel} |A| = 4

STEP 2 — EXTRACT SUMMARY ENTITY SET S per model

TF-IDF (verbatim copy, original casing preserved):

Summary text retains "Dr. Sarah Chen", "MIT", "Earth", "James Webb"

S_tfidf = {Sarah Chen, MIT, Earth, James Webb Space Tel} |S| = 4

BiLSTM (BPE lowercased output):

Summary: "dr chen at mit led study using webb telescope..."

"dr chen" -> lowercase -> spaCy rule misses PERSON

"MIT" stays all-caps -> detected as ORG

"earth", "james webb" -> lowercase -> not detected

S_bilstm = {MIT} |S| = 1

Transformer (strict BPE lowercase):

Summary: "scientists found a huge black hole close to earth. mit team..."

"earth" lowercase -> not GPE "mit" lowercase -> not ORG

"dr chen", "james webb" lowercase -> not detected

S_trans = {} |S| = 0 (casing artifact, NOT a factual error)

STEP 3 — COMPUTE P = |S ∩ A|/|S| R = |S ∩ A|/|A| F1 = 2PR/(P+R)

TF-IDF: |S ∩ A| = 4

P=4/4=1.00 R=4/4=1.00 F1=1.00 (perfect — verbatim copy preserves all)

BiLSTM: |S ∩ A| = 1 (only MIT)

P=1/1=1.00 R=1/4=0.25 F1=2*1*0.25/1.25 = 0.40

Transformer: |S|=0 so |S ∩ A|=0

P=1.00 (convention: 0 hallucinations) R=0.00 F1=0.00 <- BUG

THE CASING BUG — evaluation artifact, not model failure

BPE vocabulary built on lowercased corpus -> model outputs lowercase

spaCy rule-based: "earth"!="Earth" -> not GPE; "mit"!="MIT" -> not ORG

Transformer is factually accurate — it says "earth" meaning Earth.

FIX: capitalise first char per sentence OR use neural NER (en_core_web_lg)

NER SUMMARY TABLE

Model	S entities	|S ∩ A|	Precision	Recall	F1	
TF-IDF	4	4	1.00	1.00	1.00	
BiLSTM	1	1	1.00	0.25	0.40	
Transformer	0	0	1.00*	0.00	0.00	<- BUG

(3) SENTIMENT CONSISTENCY [EXTRINSIC]

NO REFERENCE NEEDED

CLASSIFIER: Logistic Regression on TF-IDF vectors (~5000 features)

Classes: POSITIVE | NEUTRAL | NEGATIVE trained on labeled news corpus

STEP 1 — TF-IDF VECTORISE ARTICLE (top activations)

x["scientists"] = TF*IDF = 0.111*5.45 = 0.605

x["hole"] = 0.111*5.86 = 0.651   x["telescope"] = 0.111*4.69 = 0.521

x["discovered"] = 0.111*3.50 = 0.388   x["earth"] = 0.111*4.54 = 0.504

Science/discovery words dominate the feature vector.

STEP 2 — LR SCORE z_k = W_k dot x + b_k

LR learns separate weight vector W_k per class k.

z_pos = 1.2*0.605 + 0.9*0.388 + 0.8*0.651 + 0.5*0.504 + ...

z_pos = 0.726 + 0.349 + 0.521 + 0.252 + ... = 2.02

z_neu = 0.8*0.605 + 0.6*0.388 + 0.2*0.651 + 0.3*0.504 + ...

z_neu = 0.484 + 0.233 + 0.130 + 0.151 + ... = 1.53

z_neg = -0.3*0.605 - 0.2*0.388 + 0.9*0.350 + 0.1*0.651 + ...

z_neg = -0.182 - 0.078 + 0.315 + 0.065 + ... = 0.17

STEP 3 — SOFTMAX P(k|x) = exp(z_k) / SUM_j exp(z_j)

exp(z_pos)=exp(2.02)=7.54 exp(z_neu)=exp(1.53)=4.62 exp(z_neg)=exp(0.17)=1.19

SUM = 7.54 + 4.62 + 1.19 = 13.35

P(POSITIVE) = 7.54 / 13.35 = 0.565

P(NEUTRAL) = 4.62 / 13.35 = 0.346

P(NEGATIVE) = 1.19 / 13.35 = 0.089

Article label: POSITIVE (discovery = exciting, positive framing)

STEP 4 — APPLY SAME LR TO EACH SUMMARY -> label match?

TF-IDF summary:

Same keywords: scientists, discovered, massive, black, hole, telescope

z_pos=1.82 z_neu=1.38 z_neg=0.14 -> P(POS)=0.558 label: POSITIVE

Article=POSITIVE Summary=POSITIVE -> MATCH

BiLSTM summary:

Keywords: researchers, found, large, black, hole, earth, webb

z_pos=1.61 z_neu=1.27 z_neg=0.19 -> P(POS)=0.541 label: POSITIVE

Article=POSITIVE Summary=POSITIVE -> MATCH

Transformer summary:

Keywords: scientists, found, huge, black, hole, close, earth

z_pos=1.74 z_neu=1.35 z_neg=0.11 -> P(POS)=0.554 label: POSITIVE

Article=POSITIVE Summary=POSITIVE -> MATCH

RESULTS

All 3 models: MATCH (all summaries preserve positive sentiment)

Real corpus match rates: TF-IDF 78% BiLSTM 83% Transformer 87%

KEY INSIGHT: Extrinsic eval needs NO reference summary!

A mismatch (neg tone) would mean summary distorted the meaning.

(4) TOPIC CLASSIFICATION [EXTRINSIC]

NO REFERENCE NEEDED

CLASSIFIER: Random Forest (30 decision trees, majority vote)

Classes: Politics Sports Science Technology Business Health (10 total)

Features: TF-IDF top-1000 words. Feature subset per node: sqrt(1000) ~ 32

STEP 1 — ARTICLE FEATURE VECTOR (top activations)

x["scientists"]=0.605 x["hole"]=0.651 x["telescope"]=0.521

x["discovered"]=0.388 x["earth"]=0.504 x["massive"]=0.466

Astronomy words dominate -> strong SCIENCE signal.

STEP 2 — ONE DECISION TREE NODE (entropy-based split)

Node N contains: {Science:4, Technology:3, Politics:3} n=10

H(N) = -(4/10*log2(4/10) + 3/10*log2(3/10) + 3/10*log2(3/10))

= -(0.4*-1.322 + 0.3*-1.737 + 0.3*-1.737) = 1.571 bits

Try split: "telescope" > 0.30

H(LEFT tel>0.30) = -(3/4*log2(3/4) + 1/4*log2(1/4)) = 0.811

H(RIGHT tel<=0.30) = -(2/5*log2(2/5) + 3/5*log2(3/5)) = 0.971

IG = 1.571 - (4/10*0.811 + 6/10*0.971) = 1.571 - 0.907 = 0.664 bits

Split chosen if IG > all other candidate features at this node.

STEP 3 — 30 TREES VOTE (bootstrap sampling + random feature subset)

Tree 01: [telescope=0.52, black=0.35, scientists=0.61] -> SCIENCE

Tree 02: [hole=0.65, discovered=0.39, earth=0.50] -> SCIENCE

Tree 03: [massive=0.47, galaxy=0.39, space=0.41] -> SCIENCE

Trees 04-27: astronomy features dominate all splits -> SCIENCE (24x)

Tree 28: [technology=0.28, space=0.41] -> TECHNOLOGY

Tree 29: [chen=0.35, team=0.28, university=0.15] -> SCIENCE

Tree 30: [billion=0.30, sun=0.25, formed=0.18] -> SCIENCE

Vote tally: SCIENCE=28 TECHNOLOGY=2 -> Majority: SCIENCE

STEP 4 — SAME RF ON EACH SUMMARY -> label match?

Article label: SCIENCE

TF-IDF: [scientists,discovered,telescope] dominant

RF vote: SCIENCE=25/30 label: SCIENCE -> MATCH

BiLSTM: [black,hole,earth,webb,telescope] kept

RF vote: SCIENCE=22/30 label: SCIENCE -> MATCH

Transformer: [found,huge,black,hole,close,earth] kept

RF vote: SCIENCE=24/30 label: SCIENCE -> MATCH

RESULTS

All 3 models: MATCH (all summaries preserve Science topic)

Real corpus 5-fold CV: 86.2% +/- 1.8%

Match rates: TF-IDF 81% BiLSTM 85% Transformer 89%

KEY INSIGHT: Topic mismatch = summarizer drifted to wrong subject.

STAGE 5

INTRINSIC vs EXTRINSIC EVALUATION — KEY DISTINCTION

INTRINSIC (Modules 1+2)

Compares summary against reference or article.

ROUGE: n-gram overlap with human reference.

NER: entity recall from original article.

Paraphrases score ZERO ("found"!="discovered").

Needs reference. Measures lexical fidelity.

EXTRINSIC (Modules 3+4)

Tests summary usefulness on downstream task.

Sentiment: does summary preserve tone?

Topic: does summary preserve subject?

No reference needed. Paraphrases PASS.

Measures real-world utility, not surface text.

METRICS SCORECARD — PER-MODEL COMPARISON

	TF-IDF EXTRACTIVE	BiLSTM+BAHDANAU	TRANSFORMER BEST
ROUGE-1 F1:	17.0%	~20%	24.5%
ROUGE-2 F1:	6.1%	~8%	9.8%
ROUGE-L F1:	9.2%	~14%	19.9%
NER Precision:	1.00	1.00	1.00
NER Recall:	1.00	0.25	0.00*
NER F1:	1.00	0.40	0.00*
Sentiment Match:	78%	83%	87%
Topic Match:	81%	85%	89%

* Transformer NER R/F1=0: BPE lowercase -> spaCy misses "earth","dr chen". Fix: capitalise tokens or use neural NER.

STREAMLIT DASHBOARD — INTERACTIVE EVALUATION INTERFACE

N-GRAM PANEL

- Bar chart: ROUGE-1/2/L
- All 3 models compared
- Relative gain labels
- +117% on ROUGE-L
- P / R / F1 per metric
- Download CSV

NER PANEL

- Entity table per model
- PERSON ORG GPE FAC
- Highlight missed entities
- P/R/F1 bars
- Casing bug flag
- Article entity list

SENTIMENT PANEL

- Pie: POS / NEU / NEG
- Article vs Summary
- Match indicator
- Softmax probability bars
- z-score per class
- LR weights heatmap

TOPIC PANEL

- Predicted topic chip
- Confidence bars
- Tree visualization
- 30-tree vote histogram
- 5-fold CV: 86.2%
- Feature importance